

NAD⁺ (Nicotinamide Adenine Dinucleotide)

Generic Name:

Nicotinamide Adenine Dinucleotide (NAD⁺)

Synonyms:

- NAD⁺
 - NADH (reduced form)
 - β-Nicotinamide Adenine Dinucleotide
 - Coenzyme I
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Brand Names (compounded formulations):

- NAD⁺ IV Therapy
- NAD⁺ Drip
- Rejuvenation NAD

Note: No FDA-approved branded formulations; all use is off-label and compounded.



Lifestyle Compounding Pharmacy
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Formulations and Dosage (varies by provider):

Form	Typical Dose Range	Route
NAD ⁺ IV infusion	100–1000 mg per session	IV infusion (slow drip)
NAD ⁺ IM injection	50–250 mg	Intramuscular
NAD ⁺ oral supplement	100–500 mg/day	Oral (low bioavailability)
NAD ⁺ nasal spray or sublingual	10–50 mg	Mucosal (experimental)

IV infusions are typically given over 1–4 hours depending on the dose.

Mechanism of Action:

NAD⁺ is a coenzyme found in all living cells. It plays a central role in:

- **Energy production** via the mitochondrial electron transport chain (oxidative phosphorylation)
- **DNA repair** via activation of PARPs (poly ADP-ribose polymerases)
- **Gene expression and longevity pathways**, particularly via activation of sirtuins (SIRT1–7)
- **Neuroprotection** and cell signaling



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Proposed Indications (Off-Label / Investigational):

Note: These uses are not FDA-approved and are based on clinical observation, anecdotal evidence, or ongoing research.

- Chronic fatigue syndrome
 - Substance use detoxification (especially alcohol, opioids, benzodiazepines)
 - Cognitive decline / neurodegenerative disorders (e.g., Alzheimer's, Parkinson's)
 - Depression and anxiety (adjunctive)
 - Aging-related mitochondrial dysfunction
 - Athletic performance and recovery
 - Sleep disturbances
 - General cellular rejuvenation / anti-aging
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Contraindications:

- Known hypersensitivity to NAD⁺ or formulation components
- Active malignancy (theoretical concern due to enhanced metabolic activity)
- Severe liver or kidney disease (use with caution)



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Precautions:

- **IV infusion rate** must be carefully controlled to prevent adverse reactions
 - Patients may experience discomfort during rapid infusions
 - Use with caution in individuals with heart disease due to vasodilation effects
 - Ensure hydration status is adequate during infusion
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Adverse Effects:

Common:

- Flushing or warmth
- Chest or abdominal tightness
- Nausea
- Lightheadedness
- Anxiety or restlessness during infusion

Less Common:

- Palpitations
- Headache
- Diarrhea

Rare/Severe:

- Hypotension (with rapid IV administration)
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- Injection site inflammation (with IM or IV)
 - Allergic reaction (rare)
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Drug Interactions:

- **Chemotherapeutics** (theoretical concern with cell metabolism)
- **MAO inhibitors / antidepressants** (potential for interaction in neuropsychiatric patients)
- **Antioxidants** – Synergistic effects possible

No major known interactions with most commonly prescribed medications.

Monitoring Parameters:

- Vital signs during and post-infusion (heart rate, BP)
 - Electrolytes in high-dose or repeated use
 - Mental status in psychiatric or detox protocols
 - Liver and kidney function in long-term use
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Pregnancy Category:

Not classified by FDA. Avoid use unless under physician supervision.



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Storage and Stability:

- Store at 2–8°C (refrigerated)
 - Protect from light
 - Discard unused portion after opening
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Clinical Considerations:

- Effects are often **dose- and rate-dependent** — slow infusion is critical for tolerability.
- Some clinics use NAD⁺ as part of **detox protocols**, combined with amino acids or micronutrients.
- Increasing evidence suggests NAD⁺ may support **mitochondrial health**, **neuroplasticity**, and **longevity**, but high-quality human trials are limited.